

Mech support\test4.cpp

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1  /*****
2
3  Q1. Write a C++ program for an autonomous line-following robot with 8 IR sensors arranged from left
   to right.
4
5  Each sensor gives:
6  0 → White surface
7  1 → Black line
8
9  Example sensor input:
10 0 0 1 1 1 0 0 0
11
12 Requirements:
13 1. Store the 8 sensor readings in an array.
14 2. Take all 8 sensor readings as input from the user.
15 3. Calculate the position of the detected line using the sensor indices.
16 4. If the line is detected more toward the left, print "Turn Left".
17 5. If the line is detected more toward the right, print "Turn Right".
18 6. If the line is centered, print "Move Forward".
19 7. If no sensor detects the line, print "Line Lost".
20 8. Create separate functions for:
21     - Reading the sensor values
22     - Calculating the line position
23     - Deciding the robot's movement
24 9. Do not use a separate if-else condition for every possible sensor combination.
25
26 Example:
27
28 Input:
29 0 0 1 1 1 0 0 0
30
31 Output:
32 Line Position: Center
33 Action: Move Forward
34
35 code by Rayan
36 *****/
37 #include <iostream>
38 #include <string>
39 using namespace std;
40 int ir_arr[8];
41 int total_pos = 0;
42 int total_sensors = 0;
43 int reading_flag = false;
44 string line_pos;
45 string action_dir;
46 void read_sensors(){
47     for(int i = 0 ; i <8;i++){
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48     cout << "Enter the sensor reading for sensor no."<<i+1<< ":";
49     cin >> ir_arr[i];
50 }
51 }
52
53 int calc_line(){
54     for(int i = 0; i < 8;i++){
55         if(ir_arr[i]==1){
56             total_pos+=i;
57             total_sensors++;
58             reading_flag = true;
59         }
60     }
61     if(reading_flag == false){
62         return -1;
63     }else{
64         return (double) total_pos/total_sensors;
65     }
66 }
67
68 int plot_pos(double pos_ave){
69     if(pos_ave==3){
70         line_pos = "Center";
71         action_dir = "Move Forward";
72     }else if(pos_ave>3){
73         line_pos = "Right";
74         action_dir = "Turn Right";
75     }else if(pos_ave<3 && pos_ave >0){
76         line_pos = "Left";
77         action_dir = "Turn Left";
78     }else if(pos_ave == -1){
79         line_pos = "None";
80         action_dir = "Line Lost";
81     }
82     cout << "Line Position: "<< line_pos<<endl;
83     cout << "Action: "<<action_dir<< endl;
84 }
85
86 int main()
87 {
88     read_sensors();
89     double pos_ave = calc_line();
90     plot_pos(pos_ave);
91
92
93     return 0;
94 }
```